

Byron Hodel

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October 31st, 2025

To the Hiring Manager,

I saw your posting for the position of Applications Programmer in the College of Engineering Robotics. I feel that my background aligns with the required qualifications and core responsibilities outlined for the position.

I recently graduated with a B.S. degree in Software Engineering and CIS Math from the University of Michigan-Dearborn. This has given me a solid foundation that I hope to build by career. My many years of experience as a Peer Tutor at Washtenaw Community College has provided me with experience working with a variety of people in an education setting.

I have extensive skills in programming using common languages, including Python, C, C++, C#, Java, and others. I feel that I have the ability to program in the capacity required for this role. I'm looking to begin software development professionally and my project experience demonstrates proficiency in tasks similar to those described in the job posting:

- **Experience working in higher education:** I currently serve as a Peer Tutor at Washtenaw Community College, a role you have held since October 2019, demonstrating my experience in an academic setting.
- **Building custom tools:** My skills include proficiency in both Python and Javascript, along with a variety of other languages. This knowledge and experience, highlighted in my personal projects, illustrate my ability to develop custom tools.
- **Analysis and Requirement Gathering:** My strong communication skills, developed while tutoring, align with the need to assist in gathering and reviewing requirements and working with non-technical stakeholders.

Sincerely,

Byron Hodel

Byron Hodel

SOFTWARE ENGINEER

750 Drawin Dr. Chelsea MI

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Summary

I am a recent graduate from the University of Michigan-Dearborn looking to begin software development professionally. I have a lot of experience programming in both low level and high level languages such as C, C++, C#, Java, Python, etc. I am capable and willing to learn new technologies and APIs to further my engineering and problem solving skills.

Education

University of Michigan-Dearborn

B.S. IN SOFTWARE ENGINEERING AND CIS MATH

Dearborn MI

Aug. 2022 - Aug. 2025

Washtenaw Community College

ASSOCIATES OF SCIENCE

Ann Arbor MI

Jan. 2019 - May. 2022

Skills

Languages C, C++, Odin-lang, C#, Java, Python, Zig, CMake, Javascript, HTML+CSS

Software Windows, Linux, MySQL, GDB

Experience

Washtenaw Community College

PEER TUTOR

Ann Arbor, MI

Oct. 2019 - Present

- Tutored students in the fields of Mathematics and Computer Science.

Projects

TTF Font Rasterizer

Odin, Raylib

A MINIMAL TTF PARSER THAT EXTRACTS SIMPLE GLYPHS FROM A TTF FILE.

- Implemented a minimal TrueType parser and custom rasterizer to extract and render glyphs into a texture atlas.
- Built an interactive glyph inspector to visualize pixel grids and identify rasterization artifacts.

REST API Web Server

Go

- Built a simple rest API that can respond to HTTP requests with Json data.

General Purpose Memory Allocator

Odin

A SOMEWHAT SLOW, BUT FLEXABLE MEMORY ALLOCATOR

- Designed a flexible free-list allocator with block coalescing and manual heap management.
- Enhanced understanding of fragmentation and low-level memory operations.

Memory Arena Allocator

C

A LIMITED, BUT VERY FAST MEMORY ALLOCATOR

- Created a fixed-size arena allocator for temporary allocations, achieving constant-time performance.
- Provides stack-like semantics and eliminates memory leaks via full arena resets.

General Datastructure implementations

C, Zig

FROM WHEN I TOOK A DATASTRUCTURES AND ALGORITHMS COURSE

- Implemented a Dynamic Array, growing the array when it ran out of memory.
- Implemented Binary and AVL Trees.
- Implemented A Hash Map using multiple algorithms for resolving collisions.
- For each of the above algorithms, I created tests to validate functionality and created code to collect statistics. For example, how many rotations the AVL tree has done to maintain balance